

# Applying Exponent Laws

*Teacher's Edition*

## The Period

**MA.8.NSO.1.3** Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

**Must** Rational-number bases with integer exponents: rewrite with positive exponents, and apply several laws to generate an equivalent expression.

**Must not** No fractional exponents. Bases are nonzero, and the exponent's sign never moves the base's sign.

**Target** I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

**MTR** **MTR.5.1** — Notes I · **MTR.4.1** — Board 5 · **MTR.6.1** — Board 3

| Segment                             | Slides | Time          |
|-------------------------------------|--------|---------------|
| Settle in; post the learning target | 1      | 1 min         |
| Warm-Up                             | 2–3    | 5 min         |
| Notes                               | 4–6    | 11 min        |
| Example 1                           | 7–8    | 3 min         |
| Your Turn                           | 9–10   | 3 min         |
| Example 2                           | 11–12  | 3 min         |
| Your Turn                           | 13–14  | 3 min         |
| Whiteboards                         | 15–32  | 13 min        |
| Independent Set                     | 33     | 6 min         |
| IXL                                 | 34     | 5 min         |
| <b>Total</b>                        |        | <b>53 min</b> |

### Say these three things out loud today

1. A negative exponent flips the base. On a fraction, that means turn it over.
2. The base keeps its own sign through the flip. Negative two-thirds flips to negative three-halves, not three-halves.
3. A whole number is a fraction over 1, so  $4^{-3}$  and  $(1/4)^3$  are one rule, not two.

## Slide by Slide

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1. **Applying Exponent Laws.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes..** Q4 is today's move: a negative fraction's reciprocal is negative.
3. **Warm-Up — Answers..** Reveal. 1.  $\frac{1}{8}$  2.  $7^2$  3.  $3^4$  ( $81 > 64$ ) 4.  $-\frac{5}{2}$
4. **Notes — Copy the table..** To divide by two fifths you multiply by five halves. That is why the base flips.
5. **Notes.** Reciprocal flips top and bottom. It does not touch the sign.
6. **Notes.** Fernanda: quotient of powers first, then power of a power, then rewrite the decimal as a fraction and flip.
7. **Example 1 — Flip, then evaluate..** SPOKEN ONLY.
8. **Example 1 — Worked..** Flip:  $(-8/7)^2$ .
9. **Your Turn — Same steps, your numbers. Boards up when done..** Catches a student who flips the sign with the fraction (the answer here is negative, and a board with  $+8/125$  dropped it) and one who never flipped ( $-125/8$ ).
10. **Your Turn — Answer..** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — Two laws, then a flip..** SPOKEN ONLY.
12. **Example 2 — Worked..** Product of powers:  $-2 + 4 = 2$ .
13. **Your Turn — Same steps, your numbers. Boards up when done..** Catches a student who adds to  $-1$  and stops there (a negative exponent is not a finished answer), and one who writes  $(5/4)^{10}$  (added  $7 + 3$  but kept the minus).
14. **Your Turn — Answer..** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** A recipe is scaled by a factor of  $(2/3)^{-4}$ . The original recipe serves 6 people. Rewrite the factor with a positive exponent.  $\rightarrow \left(\frac{3}{2}\right)^4$  *The number of people it serves is not needed.  $(2/3)^4$  — the minus was dropped without flipping [8.NSO.1.3 negative exponent law];  $-(3/2)^4$  — negative exponent read as a negative value [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]; anything built from 6 — the unneeded figure was used*
- 17-18. **Board 2.**  $\left(\frac{3}{-5}\right)^{-9} \rightarrow \left(\frac{-5}{3}\right)^9$   $(5/3)^9$  — took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]
- 19-20. **Board 3.**  $\left(-\frac{8}{9}\right)^{-2} \rightarrow \frac{81}{64}$   $-81/64$  — sign kept through an even power;  $64/81$  — never flipped [8.NSO.1.3 negative exponent law]
- 21-22. **Board 4.**  $\left(\frac{21}{4}\right)^{-5} \rightarrow \left(\frac{4}{21}\right)^5$   $(21/4)^5$  — minus dropped without flipping;  $4/21^5$  — flipped but applied the exponent to the denominator only
- 23-24. **Board 5.**  $\left(-\frac{7}{8}\right)^{-3}$  Which expression is equivalent?  $\rightarrow \mathbf{A - (-8/7)^3}$  (B) took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]; (C) dropped the minus on the exponent without flipping [negative exponent law not applied]; (D) took the opposite of the base instead of its reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.
- 25-26. **Board 6.**  $\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$  Which expression is equivalent?  $\rightarrow \mathbf{A - (5/8)^2}$  (B) flipped a base whose exponent was already positive [negative exponent law applied]

where no negative exponent remained]; (C) added  $-2 + (-4)$  instead of subtracting [quotient law: subtract the exponents]; (D) multiplied  $-2 \cdot -4$  [confused with power of a power] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**27-28. Board 7.**  $\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1 \rightarrow \frac{25}{4}$  1 — treated  $(2/5)^{-1}$  as  $2/5$  and multiplied by its reciprocal [negative exponent law not applied];  $5/2$  — only one factor counted

**29-30. Board 8.**  $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3 \rightarrow \left(\frac{2}{3}\right)^9$   $(2/3)^6$  — added  $3 + 3$  in the power-of-a-power step [confused with product of powers];  $(3/2)^9$  — flipped a positive exponent

**31-32. Board 9.** Marco says  $(-2/3)^{-5}$  can be rewritten as  $(2/3)^5$  because 'two negatives make a positive.' Is he right?  $\rightarrow$  **No** —  $(-2/3)^{-5} = (-3/2)^5$  'Yes' — opposite taken as well as reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]

**33. Independent Set — Six questions. On your own, in silence..** Hand out the Independent Set.

**34. IXL — Last five minutes..** IXL: every listed skill is required, SmartScore 67, due at the start of the next class.

## Misconceptions to Watch

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**Board 1.**  $(2/3)^4$  — the minus was dropped without flipping [8.NSO.1.3 negative exponent law]

**Board 2.**  $(5/3)^9$  — took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]

**Board 3.**  $-81/64$  — sign kept through an even power

**Board 4.**  $(21/4)^5$  — minus dropped without flipping

**Board 5.** took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]

**Board 6.** flipped a base whose exponent was already positive [negative exponent law applied where no negative exponent remained]

**Board 7.** 1 — treated  $(2/5)^{-1}$  as  $2/5$  and multiplied by its reciprocal [negative exponent law not applied]

**Board 8.**  $(2/3)^6$  — added  $3 + 3$  in the power-of-a-power step [confused with product of powers]

**Board 9.** 'Yes' — opposite taken as well as reciprocal [B1G-M 8.NSO.1.3 task:  $-b$  vs  $b^{-1}$ ]

**Materials** Whiteboards and markers. Calculators are allowed on everything;  $(-7/8)^{-2}$  on a calculator returns 1.306..., which is  $64/49$  — worth converting once so students trust both forms.